



INTERIM REPORT

Higher Diploma in Science in Computing

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Stream

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1. Introduction

Reusable product containers play a critical role in pharmaceutical manufacturing and must be carefully managed to ensure product quality, traceability, and compliance with Good Manufacturing Practice (GMP) requirements. Before a container can be introduced into production, it must go through a controlled cleaning process and receive Quality Assurance (QA) approval confirming its suitability for production use. Once approved, the container enters a production lifecycle where it may be used for a maximum of fourteen production runs or thirty calendar days, whichever comes first. When either limit is reached, the container must be removed from production, cleaned, and approved by a supervisor before beginning a new lifecycle.

Traditionally, these activities have been recorded using paper-based logbooks, making the process susceptible to human error, incomplete records, and reduce traceability. Manual systems also make it more difficult to monitor container status, enforce lifecycle rules, and demonstrate compliance during audits.

This project addresses these challenges through the development of a full-stack Container Management System that digitises the complete container lifecycle. The application records container movements, manages QA and Supervisor approval workflows, enforces lifecycle limits, and provides operational dashboards that improve visibility of container activity. A simulation engine has also been developed to generate realistic operational data for testing and analytics.

2. Project Scope and Objectives

The scope of the project is to design and develop a full-stack web application that manages the complete lifecycle of reusable production containers used in pharmaceutical manufacturing. The system provides end-to-end traceability by recording container movements, enforcing lifecycle rules, and managing role based approval workflows. The application supports users throughout the container lifecycle, from initially being received and cleaned through Quality Assurance (QA) approval, production use, lifecycle expiry, cleaning, supervisor approval, and reintroduction into production. In addition to workflow management, the application provides operational dashboards and analytic to support monitoring and decision-making.

The project scope includes:

- User authentication using JSON Web Tokens (JWT)
- Role-based access control
- Container lifecycle management
- Workflow validation and business rule enforcement
- QA approval workflow
- Supervisor approval workflow
- Container movement tracking
- Operational dashboard with key performance indicators
- Analytics dashboard with data visualisation
- Quick container scan and search functionality
- Simulation engine for generating operational datasets
- Unit testing using Jest

The objectives of the project are to:

- Design and implement a relational MySQL database to support container lifecycle management
- Develop a secure RESTful API using Node.js and Express
- Develop a responsive React frontend for managing containers and approvals
- Implement a workflow validation that enforces container lifecycle rules and approval requirements.
- Implement role-based authentication and authorisation using JWT
- Record container movements and approval history to provide complete traceability
- Develop dashboards that present operational metrics through interactive charts and reports
- Generate simulated operational data to support testing and analytical reporting
- Prepare operational datasets suitable for regression analysis
- Test the application to verify functionality, reliability, and compliance with business requirements
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3. Student's Learning Objectives

The learning objectives for this project were to develop practical software engineering skills while designing and implementing a real-world solution for managing production containers within a pharmaceutical manufacturing environment. The project has provided an opportunity to apply software development principles throughout the software development lifecycle, from requirements analysis and system design to implementation, testing, and documentation. It has also strengthened knowledge of modern full-stack development while demonstrating the importance of workflow automation, traceability, and compliance within regulated industries.

The learning objectives are to:

- Strengthen full-stack development skills using React, Node.js, Express, and MySQL
- Apply software engineering principles throughout the planning, design, implementation, testing, and documentations phases of the project
- Develop a secure RESTful web application using authentication, role based access control, and API development techniques
- Design and implement a relational database that's capable of supporting complex workflow and approval processes
- Gain practical experience implementing business rules and workflow validation within a real-world software application
- Develop interactive dashboards and data visualisations to present operational information using Chart.js
- Generate realistic operational datasets through simulation for testing and analytical purposes
- Gain practical experience using Git, GitHub, Thunder Client and Jest as part of a modern software development workflow
- Develop project management skill through the application of Agile Scrum principles during software development
- Improve problem solving and debugging skills by resolving technical challenges encountered throughout the implementation of the project

4. Technical Specification of the Project

The Container Management System has been developed using modern full-stack architecture that separates the frontend, backend, and database. This Improves maintainability, scalability, and security while supporting the implementation of business rules that govern the pharmaceutical container lifecycle. The system follows a client-server model, where React based frontend communicates with a RESTful API developed using Node.js and Express. The backend processes business logic, enforces workflow validation, manages authentication and authorisation, and interacts with a MySQL database for persistent data storage.

The project was developed and tested using the following hardware and software technologies:

- Windows based personal computer
- Internet connection

Software and Development Tool

- Visual Studio Code - IDE
- Node.js- JRE
- Express – backend web application framework
- React with Vite- frontend development framework
- Material UI- user interface component library
- MySQL- relational database management system
- MySQL workbench- design and administration
- Git- version control
- GitHub- repository
- Thunder Client – REST API development and testing
- Axios-client-server communication
- JWT-user authentication and authorisation
- Chart.js- dashboard visuals and analysis
- Jest- backend unit testing
- npm- package management
- Trello – Agile scrum project planning and sprint management
- Figma- user interface wireframing
- Draw.io - diagrams
- Lucid Chart- diagrams

Database

A MySQL relational database stores application data including user accounts, containers, workflow movements, approval requests, and audit information. Primary and foreign key relationships maintain data integrity while supporting complete traceability throughout the container lifecycle.

Testing

The application has been tested throughout development using a combination of manual and automated testing techniques. REST API endpoints were validated using Thunder Client, while backend functionality was tested through unit testing with Jest. This testing was carried out by validating complete workflow scenarios, including authentication, approval processes, lifecycle enforcement, and container movement tracking.

5. Project Plan

The Project was planned using an incremental Agile scrum approach, allowing the application to be developed in manageable stages. Development activities were organised using Trello to manage the product backlog, sprint planning, and task progression throughout the project. User interface concepts were designed using Figma, while Lucid chart and draw.io were used to produce system architecture diagrams, workflow diagrams a, UML diagrams, and database designs before implementation commenced.

Progress to date

Week	Planned	Current Status
Week 1	Requirements gathering, research and project planning	completed
Week 2	Database design and MySQL implementation	completed
Week 3	Backend API development testing	completed
Week 4	User authentication and workflow engine	completed
Week 5	Simulation engine for operational data	completed
Week 6	React Frontend development	completed
Week 7	Dashboard implementation and reporting	completed
Week 8	Analytics and regression preparation	In progress
Week 9	System testing, debugging and refinement	In progress
Week 10	Documentation and final project completion	In progress

6. Conclusion

The project has made significant progress towards the development of a full-stack Container Management for managing the lifecycle of reusable production containers within a pharmaceutical manufacturing environment. To date, the core functionality has been successfully implemented, including user authentication, role-based access, container lifecycle management, workflow validation, QA and Supervisor approval processes, operational dashboards, and a simulation engine for generating realistic operational data to work on,

Testing completed to date has demonstrated that the application successfully enforces business rules, records container movements, and provides complete traceability throughout the container lifecycle. The project has also strengthened practical knowledge of modern software development through the use of React, Node.js, Express, MySQL, and Agile Scrum development practices.

At the time of writing, the remaining work includes completing the analytics, performing regression analysis using the generated operational dataset, finalising the scrum documentation, further front-end testing, and completing the documentation. These activities will prepare the system for final evaluation and demonstrate the application satisfies the requirements and project objectives while providing a practical digital solution.